

Weird File

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

What could go wrong if we let Word documents run programs? (aka “in-the-clear”).

Hints

<https://www.youtube.com/watch?v=Y7IJnLGqTQ>

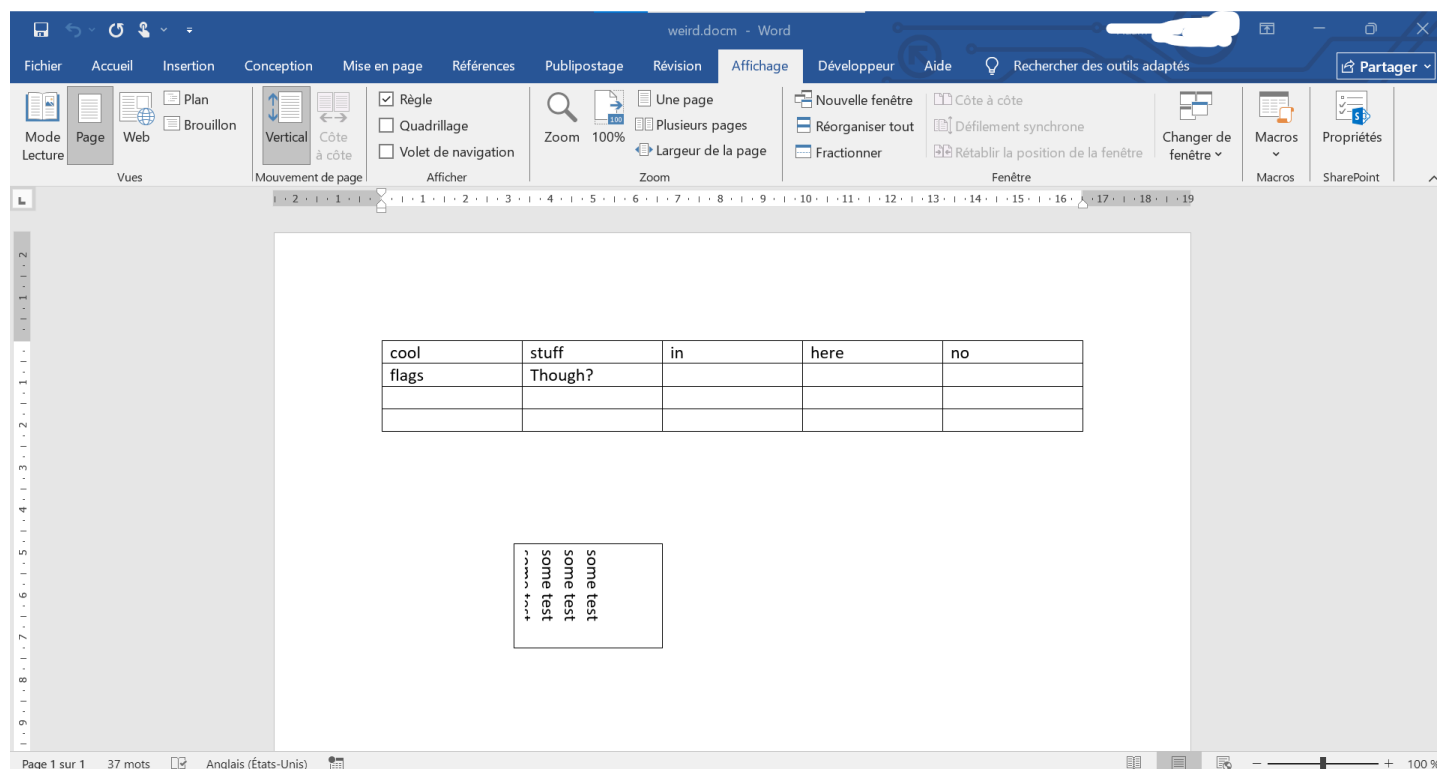
Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `unzip` – Extract ZIP archives
 - `base64` – Decode Base64-encoded content
 - **Microsoft Word** – Open and view macros
-

Complete Solution

Step 1: Download and Prepare

Download the file named `weird.docm` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type:

```
file weird.docm
```

Expected output:

```
weird.docm: Microsoft Word 2007+
```

The file is a **Word Macro-Enabled Document** (`.docm`), which is essentially a ZIP archive containing XML files and potentially VBA macros.

Step 3: Search for Visible Text

Check if the flag appears as plain text:

```
strings weird.docm | grep -i "picoCTF"
```

Output: (no results)

The flag isn't directly visible.

Step 4: Examine Metadata

Use `exiftool` to inspect metadata:

```
exiftool weird.docm
```

Output:

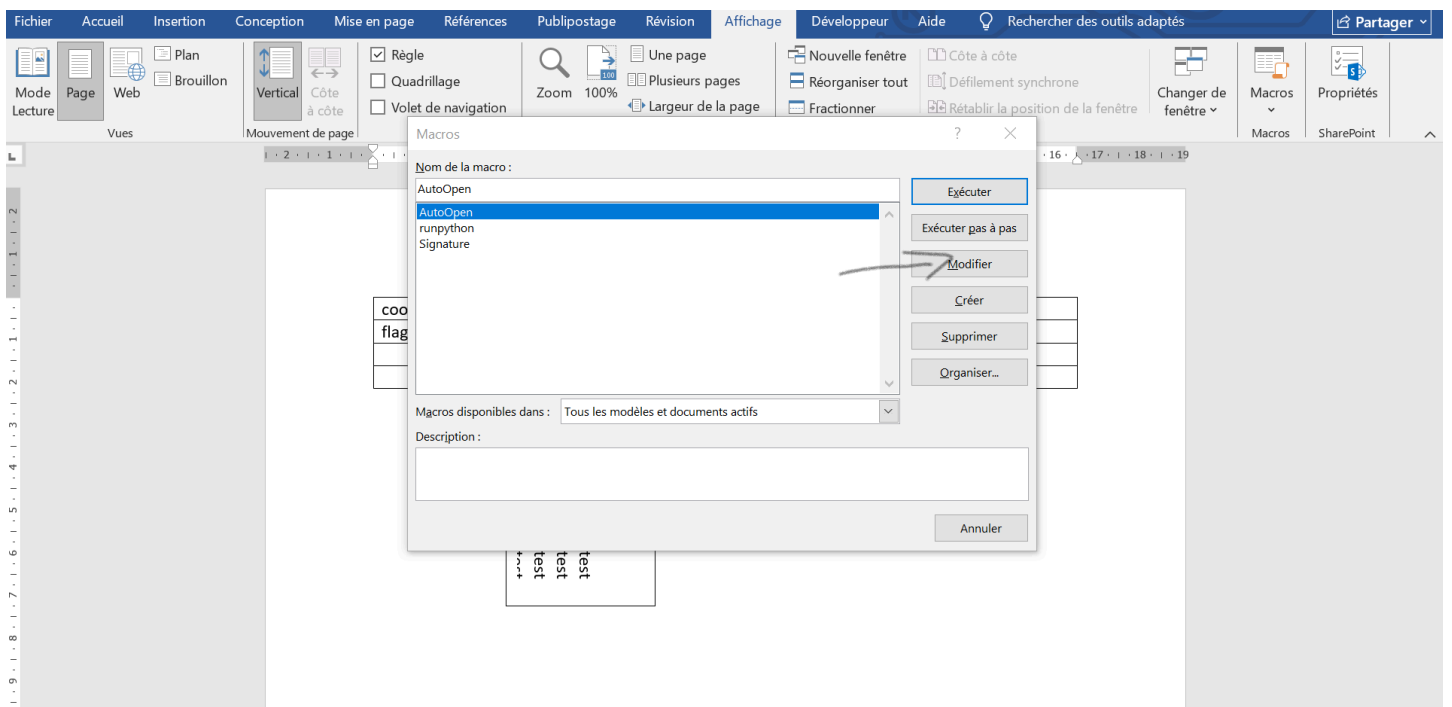
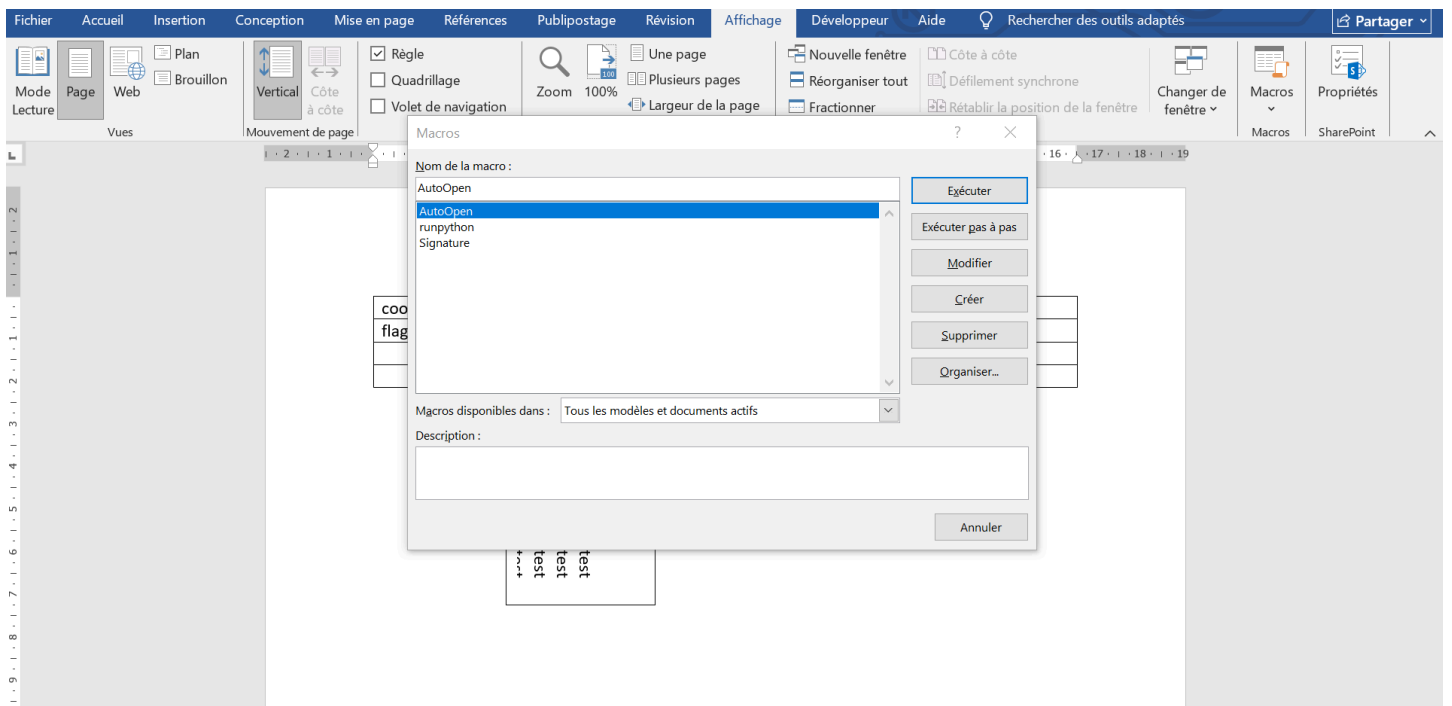
```
ExifTool Version Number      : 13.50
File Name                    : weird.docm
Directory                   : .
File Size                    : 24 kB
...
File Type                   : DOCM
File Type Extension         : docm
MIME Type                   : application/vnd.ms-word.document.macroEnabled.12
...
Create Date                 : 2020:08:10 21:54:00Z
Modify Date                 : 2020:08:10 22:16:00Z
Pages                      : 1
Words                      : 8
Application                 : Microsoft Office Word
...
```

Nothing immediately suspicious, but the presence of macros is hinted by the `.docm` extension.

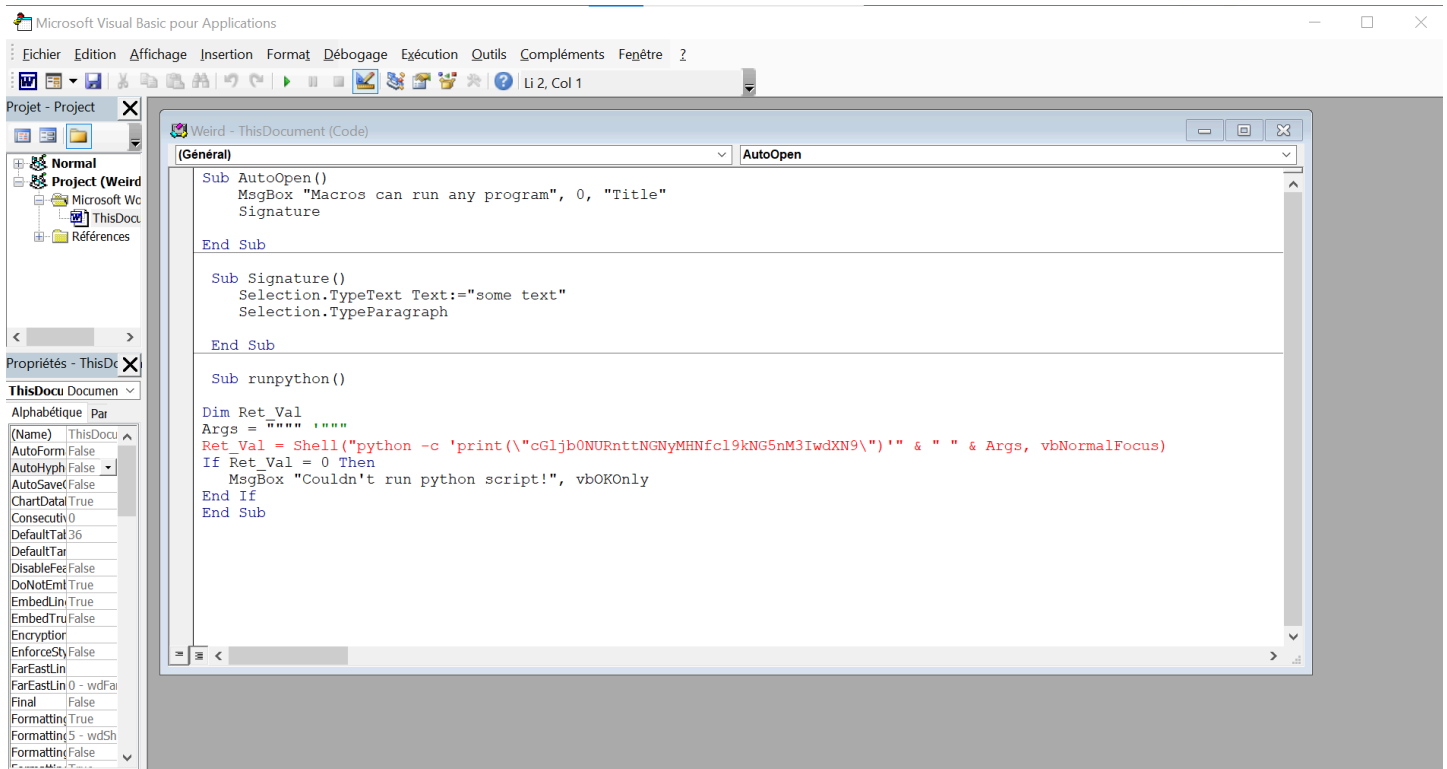
Step 5: Open the Document in Microsoft Word

The hint video demonstrates how to access macros in Word. Let's follow that approach:

1. **Open the file** in Microsoft Word
2. Go to the **View** tab
3. Click on **Macros** → **View Macros**



4. Select **AutoOpen** (a macro that runs automatically when the document opens)
5. Click **Edit**



Step 6: Examine the VBA Macro Code

The Visual Basic Editor opens, revealing the macro code:

```
Ret_Val = Shell("python -c 'print(\"cGljb0NURnttNGNyMHNfc19kNG5nM3IwdXN9\")' & \" \" & Args, vbNormalFocus)
```

Key finding: The macro executes a Python command that prints a **Base64-encoded string**!

The encoded string is: `cGljb0NURnttNGNyMHNfc19kNG5nM3IwdXN9`

Step 7: Decode the Base64

```
echo "cGljb0NURnttNGNyMHNfc19kNG5nM3IwdXN9" | base64 -d
```

Output:

```
picoCTF{<REDACTED>}
```

Step 8: Alternative Verify ZIP Header with xxd

```
xxd weird.docm | head
```

Output:

```
00000000: 504b 0304 1400 0600 0800 0000 2100 04d9  PK.....!...
00000010: 8a77 9d01 0000 3106 0000 1300 0802 5b43  .w....1.....[C
00000020: 6f6e 7465 6e74 5f54 7970 6573 5d2e 786d  ontent_Types].xm
...
```

The `PK` signature confirms this is a valid ZIP archive.

Step 9: Extract Without Opening Word

Since `.docm` files are ZIP archives, we can extract them directly without opening Word:

```
unzip weird.docm -d weird_extracted
```

Navigate to the extracted folder and look for VBA project files:

```
ls weird_extracted/word/
```

Output:

```
_rels  document.xml  fontTable.xml  settings.xml  styles.xml  theme  vbaData.xml
vbaProject.bin  webSettings.xml
```

The `vbaProject.bin` file contains the macro code. Extract strings from it:

```
strings weird_extracted/word/vbaProject.bin | grep -i "python"
```

Output:

```
Couldn't run python script!
Ret_Val = Shell("python -c 'print(\"cGljb0NURnttNGNyMHNfc19kNG5nM3IwdXN9\")'" & " " &
Args, vbNormalFocus)
```

runpython
runpython

Same Base64 string appears! Decode it as before.

✔ Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, decoding the Base64 string will reveal the complete flag.

🧠 Key Concepts

Concept	Description
DOCM Format	Word Macro-Enabled Document – a ZIP archive containing XML files and VBA macros.
VBA Macros	Visual Basic for Applications code that can run automatically when a document is opened.
AutoOpen Macro	A special macro that executes automatically when the document is opened.
Base64 Encoding	A common encoding used to represent binary data in ASCII text.
Malicious Macros	Attackers often use macros to execute commands or decode hidden payloads.

💡 Pro Tips

1. **Office documents with macros are dangerous** – `.docm`, `.pptm`, `.xlsm` files can contain executable code. Always analyze them in isolated environments.
2. **The `AutoOpen` macro runs automatically** – This is a common technique used in malware to execute code when a document is opened.
3. **Don't ignore the hint video** – The video shows exactly how to access macros in Word, which is key to solving this challenge.
4. **Extract without opening** – Use `unzip` to examine the contents without triggering any macros.
5. **`vbaProject.bin` contains the macro code** – This binary file stores the VBA project; `strings` can extract readable text from it.
6. **Base64 is often used to hide strings** – Whenever you see a long alphanumeric string with possible `=` padding, try Base64 decoding.

Alternative Base64 Decoding Tools

Tool	Command / Link
Linux <code>base64</code>	<code>echo "string" base64 -d</code>
CyberChef	https://gchq.github.io/CyberChef/ (Operation: From Base64)
Python	<code>import base64; base64.b64decode("string")</code>
Online	https://www.base64decode.org/